

Virtual Yeast for Climate-Resilient Fermentation:

A Pre-Registered Constraint-Based Map of Combined-Stress Collapse in *Saccharomyces cerevisiae*, its Co-Limitation Mechanism, and a Demonstrated Boundary of Steady-State Strain Design

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Abstract

Category: Computational Biology and Bioinformatics (CBIO). Industrial bioethanol fermentations fail when climate-driven stresses combine: heat waves, accumulating ethanol, osmotic pressure from concentrated feedstocks, and nutrient limitation interact inside the same tank, yet each stress is usually studied one factor at a time. We built a virtual *Saccharomyces cerevisiae* cell on the consensus genome-scale model yeast-GEM (4,105 reactions, 2,748 metabolites, 1,143 genes; in-repo SBML pinned by SHA-256) and extended it with four literature-anchored stress layers: cardinal-temperature enzyme kinetics, Levenspiel-type ethanol product inhibition, an obligatory osmoregulatory glycerol-production drain, and nitrogen availability scaling, with maintenance energy rising inversely with relative growth. Every functional form, parameter, environment grid, objective, and success gate was locked in a SHA-256 pre-registration before any outcome run, with a locked 70/30 design/test split of the environment space and the test region touched exactly once. Across a 5,000-environment Latin-hypercube sample of the four-axis stress space, 37.1% of environments collapse wild-type ethanol production below 20% of its reference. Bottleneck analysis of the collapsed region shows collapse is co-limited by the osmoregulatory glycerol drain (binding in 69% of solvable collapsed environments) and a heat-scaled glycolytic-capacity loss (59%), a mechanism corroborated at the regulatory level by concordance between the model's heat regime and the published heat-shock expression program (10/10 marker genes, $p = 9.8 \times 10^{-4}$). The pre-registered non-additivity gate was not met (24.2% vs. the 25% threshold; parameter-fragile under temperature-margin perturbations), and we report it as such. A pre-registered strain-design campaign - all 1,143 single-gene deletions, 435 pairs, 36 reaction scalings, a seeded evolutionary search, and a matched random baseline, evaluated once on the untouched test region - found no design that improves any locked objective beyond solver noise ($< 10^{-14}$), because collapse under this mechanism is imposed by environmental caps rather than by internal pathway limitation. Validation against three published genome-wide deletion screens likewise returns a declared null (0/4 pre-registered overlap comparisons significant), delineating a boundary of steady-state flux models for stress-tolerance engineering. All negative results are reported plainly under the pre-registration; the

principal positive contributions are the collapse map, its verified co-limitation mechanism, the regime-level transcriptome concordance, and a demonstrated method boundary for the field's most widely used engineering model class.

1 Introduction

1.1 The industrial problem: fermentation under a changing climate

Bioethanol is the largest-volume fermentation product of the industrial bioeconomy, and its production organism, *Saccharomyces cerevisiae*, operates at the edge of its physiological envelope in essentially every commercial process. A modern fuel-ethanol plant asks a yeast cell to convert a concentrated sugar mash - routinely above 200 g/L of glucose in very-high-gravity (VHG) operation - into ethanol at titers that approach the limits of its own tolerance, while the metabolic heat of fermentation and increasingly volatile ambient conditions push tank temperatures toward the top of the organism's growth range. Each of these burdens has a name and a literature. Ethanol itself is a chaotropic solvent that disorders membranes and dissipates proton gradients; its inhibition of yeast growth and fermentation has been modeled since the classical Levenspiel formulation. Temperature acts on every enzyme in the cell at once, and the cardinal-temperature model of Rosso and colleagues - a minimum, optimum, and maximum temperature with a characteristic asymmetric rise and precipitous fall of rate - has been fitted to yeast growth and fermentation kinetics across strains and conditions. Osmotic pressure, whether from the sugar of concentrated mashes or from drought-stressed feedstocks, forces the cell into an obligatory osmoregulatory program: glycerol must be synthesized and retained to balance water activity, and every mole of glycerol is carbon that cannot become ethanol. Nitrogen and other nutrient limitations shape the achievable growth rate and the maintenance burden the cell must pay before any product is made.

What the industrial literature also says, plainly, is that these stresses do not arrive one at a time. A heat wave during a VHG fermentation is a simultaneous increase in temperature, osmotic load (as sugar concentrates when water evaporates), ethanol concentration (as fermentation progresses), and nutrient stress (as the stationary phase approaches). Plant operators experience this as the "stuck fermentation": a batch that slows, stalls, and leaves residual sugar that no restart recovers. The economics are direct - a few percent of lost yield across a continental industry is measured in billions of liters - and the climate projection is worse: the frequency of multi-day heat extremes during the fermentation season is rising on every major production continent. The engineering response has so far been empirical: stress-tolerant strains are selected, evolved, or screened one stress at a time, and then combined in the hope that tolerance is additive. Sometimes it is. Often it is not, and the failures are rarely publishable, so the field's map of which combinations of stress actually break the process - and *why* - remains thin.

1.2 The scientific gap: stress is studied one axis at a time

The modeling literature mirrors the experimental one. Genome-scale metabolic models (GEMs) of *S. cerevisiae* - of which the consensus yeast-GEM is the community standard - have been used for two decades to predict gene essentiality, design metabolic-engineering interventions, and contextualize omics data. Stress, when it enters these models at all, usually enters as a single scalar

perturbation: a temperature-dependent turnover number here, an inhibitory ethanol concentration there. We are not aware of any pre-registered, genome-scale study that treats the environment as a joint, four-axis object - temperature, ethanol, osmotic pressure, and nutrient availability - and asks where in that joint space the cell's fermentation phenotype collapses, which internal constraints bind when it does, and whether the answer is robust to the chosen parameterization. The closest works study pairs of stresses, or study combined stress transcriptomically without a mechanistic flux model, or build kinetic models of a handful of reactions whose scope cannot reach the whole-cell question. The result is that the most basic map an engineer would want - here is the region of stress space where the process fails, and here is what fails first - does not exist in a reproducible, falsifiable form.

That gap is not only industrial. It is a clean instance of a general question in systems biology: when a cell fails under combined perturbations, is the failure merely the product of independent single-stress effects, or does the combination create emergent failure modes that no single-stress study can see? And if emergence exists, is it a property of the biology or of the mathematics used to describe it? We treat these as separate, testable questions, and we built the study so that each gets its own pre-registered gate.

1.3 What this paper does

We construct a virtual *S. cerevisiae* cell for combined-stress fermentation and subject it to a fully pre-registered analysis program. The model extends the consensus yeast-GEM with four stress layers, each anchored to published kinetic or physiological data and each carrying its provenance: cardinal-temperature (CTMI) scaling of enzyme-catalyzed rates; Levenspiel-type ethanol inhibition of growth-linked processes; an obligatory glycerol-production drain proportional to external osmolarity; and a nitrogen-availability scaling of nutrient uptake. Maintenance energy demand rises as relative growth falls, in a declared, assumed functional form that is never fitted to any outcome. The environment space is a locked four-axis grid - 28–42°C, 0–14% v/v ethanol, 0–1.5 M sorbitol-equivalent osmolarity, 5–100% of standard nitrogen - and the objectives are the four an industrial process cares about: ethanol yield, temperature tolerance, ethanol tolerance, and low nutrient requirement. Growth is treated as a means, never as the objective.

The analysis program is staged and gated. A1 maps single-stressor dose responses. A2 maps the combined-stress collapse surface over a 5,000-environment Latin-hypercube sample. A3 tests, against a pre-registered threshold, whether combined-stress failure is sub-additive - worse than the product of the single-stress effects - and whether that conclusion survives parameter perturbation. A4 identifies, with shadow prices and flux-variability analysis, which constraints bind in the collapsed region. A5 is a strain-design campaign under a pre-registered gate: all single-gene deletions, a seeded pair screen, reaction-bounds scalings, an evolutionary search, and a matched-budget random baseline, with final claims evaluated once on the untouched test region. A6 validates the model's stress-conditional gene-sensitivity predictions against three published genome-wide deletion screens with pre-registered overlap statistics, plus qualitative checks against two further studies. A7 asks whether the model's stress regimes are concordant with the published environmental-stress expression program at the level of marker gene sets. A8 lifts the steady-state assumption with dynamic heat-wave simulations, asking whether collapse timing reflects adaptation or accumulation.

Every gate is reported as it landed. Two of the four principal gates did not meet their pre-

registered thresholds, and an external judge round, archived verbatim with full provenance, returned a fail verdict at the program’s highest target level. We present these outcomes not as procedural embarrassments but as results: the negative theorems of this paper - about what combined-stress collapse is not, and about what steady-state strain design cannot do - are as much a part of the scientific record as the positive map and mechanism, and they are established here with the same evidentiary discipline.

1.4 Summary of principal findings

1. **A reproducible collapse map.** In the locked environment space, 37.1% of sampled combined-stress environments drive wild-type ethanol production below 20% of its unstressed reference; the collapse region has a definite geometry in the temperature–ethanol plane that the osmotic axis modulates and the (declared, see below) nitrogen axis does not.
2. **A verified co-limitation mechanism.** Collapse is co-limited by the obligatory osmoregulatory glycerol drain and by heat-scaled loss of glycolytic capacity: the two constraints bind, respectively, in 69% and 59% of solvable collapsed environments, with flux-variability analysis direction-consistent. This is the paper’s central mechanistic claim, and it is stated with its shadow-price evidence and its regime structure.
3. **Regime-level regulatory concordance.** The model’s heat regime is concordant with the published heat-shock expression program at 10/10 marker genes ($p = 9.8 \times 10^{-4}$); the nitrogen-catabolite and glycerol regimes trend or are non-significant; the glycolytic regime is declared non-concordant ($p = 0.42$). Concordance is reported per regime, not averaged into a single flattering number.
4. **A declared boundary of steady-state strain design.** Under the verified mechanism, no deletion, pair, scaling, or evolved design improves any locked objective beyond solver noise ($< 10^{-14}$ relative yield) on the untouched test region: collapse is imposed by environmental caps, not by an internal bottleneck a knockout can relieve. The model correspondingly fails to recover deletion-screen stress sensitivity (0/4 pre-registered overlap tests). Both results are reported as limitations of the model class, not of the analysis.
5. **Dynamic corroboration.** Dynamic heat-wave simulations show collapse timing is set by endogenous ethanol accumulation rather than by adaptive memory: no hysteresis beyond ethanol carry-over, and dynamic and static collapse boundaries agree.

1.5 Scope, honesty discipline, and declared limitations

This is a computational study. No wet-lab claim is made anywhere. All validation is against previously published experimental datasets; all parameters are locked before outcome runs; all parameter perturbations are one-at-a-time sweeps with declared ranges; exploratory analyses are labeled as such and never mixed with confirmatory claims. Two structural limitations are declared up front because they condition the reading of every result: first, the nitrogen lever of the environment mapping is inert in the locked rich base medium (nitrogen is supplied by amino-acid uptakes and ammonium is excreted, so scaling the ammonium bound changes nothing), which means the

nitrogen axis of the environment grid is effectively unperturbed throughout; we root-caused this during validation and report it rather than retrofitting the mapping after outcomes. Second, steady-state flux balance cannot see regulation, damage, or repair; where the paper’s questions touch those processes we say so and rely on the dynamic and transcriptome-concordance arms rather than on flux predictions. The paper’s claims are exactly the ones that survive these limitations, and they are stated in Section ?? with their evidence.

The remainder of the paper is organized as follows. Section 2 situates the work in the stress-fermentation and genome-scale modeling literatures. Section ?? gives the model, the stress layers with their equations and provenance, the environment space, the analysis stages, the gates, the datasets, and the pre-registration protocol. Section ?? reports the staged results in order. Section ?? interprets the mechanism, the negatives, and the method boundary, and states the external judge verdicts and what they changed. Section ?? concludes. Appendices carry the locked pre-registration record, parameter tables, solver-timeout registers, run logs, and the verbatim judge archive.

2 Related work

2.1 Ethanol inhibition and the kinetics of product toxicity

The quantitative study of ethanol’s effect on yeast fermentation is older than genome-scale modeling by half a century. The inhibitory effect of ethanol on growth and fermentation was formalized in the Levenspiel framework, in which the specific rate declines with product concentration according to a power law whose exponent and critical concentration are fitted per strain and condition. Classical chemostat and batch studies established that biomass yield declines steeply across the 0–107 g/L ethanol range and that maintenance requirements rise as viability falls. These studies supply two facts this paper builds on directly: ethanol inhibition is smooth and dose-dependent rather than threshold-like across most of the industrial range, and its physiological signature couples growth, fermentation rate, and viability rather than acting on any one of them alone. What the classical literature cannot supply is a whole-cell mechanism: the models are intentionally phenomenological, and their parameters move when the background medium, strain, or stress combination changes. Our treatment follows the same functional tradition - a Levenspiel-type inhibition mapped onto growth-coupled flux bounds - but embeds it in a genome-scale stoichiometric model so that its interaction with carbon drain and temperature can be read out mechanistically at the level of individual constraints.

2.2 Temperature kinetics: from Arrhenius to cardinal models

The temperature dependence of microbial rates has been described at several levels of resolution. Arrhenius-style laws capture the rising limb of the rate–temperature curve but fail at the denaturation edge; the cardinal-temperature model with inflection (CTMI) of Rosso and colleagues repairs this with three interpretable parameters - minimum, optimum, and maximum temperatures - joined by a smooth asymmetric curve that has been fitted to yeast growth and fermentation repeatedly, including systematic studies of wine and industrial strains across the full viable range. Salvado and co-workers extended such fits to mixed populations and to fermentation-relevant rates rather than

growth alone. For a whole-cell model, the CTMI form has a decisive advantage over a raw Arrhenius law: it can be applied as a single dimensionless multiplier to enzyme-catalyzed capacity while remaining honest about its aggregate character. We adopt exactly that mapping, with parameters taken from the published yeast fits and locked before any run, and we treat the maintenance-energy response to falling relative growth as a separate, declared assumption - the cell pays more to stay alive as conditions worsen - because the published evidence for the direction of that effect is strong and for its exact form weak. The sensitivity of every temperature-driven conclusion to the CTMI margin parameters is reported in Section ?? as a first-class result, since it proved to be the fragile axis of the non-additivity claim.

2.3 Osmotic stress and the cost of glycerol

The osmoregulatory program of *S. cerevisiae* is among the best-characterized stress responses in any eukaryote: hyperosmotic shock activates the high-osmolarity glycerol (HOG) pathway, induces glycerol production through GPD1/GPD2, and closes the Fps1 aquaglyceroporin to retain the polyol. Hohmann's reviews place the carbon cost of this program at the center of osmotic physiology: glycerol is not a by-product here but a requirement, and its synthesis diverts flux from glycolysis and, relevantly for industry, from ethanol. Fermentation-science studies of very-high-gravity mashes have repeatedly associated osmotic load with sluggish and stuck fermentations and with altered glycerol-ethanol partitioning. The modeling tradition, however, has rarely charged this cost explicitly to the cell in flux terms. Our osmotic layer does exactly that: an obligatory glycerol-production demand proportional to external osmolarity, parameterized from published osmotic physiology and locked, so that the carbon drain appears in the model as a constraint that can bind, carry a shadow price, and be counted - which is precisely what makes the A4 bottleneck analysis of Section ?? possible.

2.4 Combined stress in fermentation science

The industrial fermentation literature has long acknowledged combined stress without mapping it. Reviews of VHG bioethanol operation list heat, ethanol, osmotic, and nutrient stress as the canonical burden set and document that strains selected for one tolerance often disappoint under the others. Systematic multi-stress tolerance panels exist for a handful of strains and conditions, and adaptive-laboratory-evolution campaigns have produced multiply tolerant isolates whose mechanisms are usually resolved only partially and after the fact. Two features of this literature motivate the present study. First, its evidence is overwhelmingly one-factor-at-a-time at the mechanism level: even when strains are challenged jointly, the analysis attributes failure to a single dominant stress rather than to an interaction topology. Second, its computational counterpart is thin: genome-scale models have been applied to ethanol tolerance mostly through gene-essentiality or flux-coupling analyses in single stress conditions. The joint, pre-registered mapping of collapse across a four-axis stress space, with mechanism attribution and validation against published deletion screens, appears to be absent from the literature, and it is the hole this paper aims to fill - including, explicitly, the parts of the hole we could not fill and now mark on the map.

2.5 Genome-scale models, their validation culture, and this paper’s stance

The consensus yeast-GEM used here is the product of an unusually open community process, with versioned releases and a pinned SBML artifact that we carry in the repository with its hash recorded. Flux balance analysis on such models has a well-known strengths ledger - constraint accounting at genome scale, cheap and reproducible predictions, direct mechanistic readouts through shadow prices and variability analysis - and an equally well-known list of blind spots: no regulation, no damage or repair dynamics, and a steady-state assumption that is at best an averaging device under stress. The validation culture of the field has historically leaned on gene-essentiality benchmarks, which our own prior work examined critically and which this study deliberately complements with stress-conditional deletion screens as the validation target. The choice proved consequential: the pre-registered overlap tests against those screens return a null, and that null, reported plainly in Section ??, is one of the paper’s results rather than a failed preliminary. A growing methodological literature argues that negative results of exactly this kind - where a widely used model class fails a fair, pre-registered test - are necessary for the field’s self-correction; this paper is written in that tradition.